



The Toronto Nanofabrication Centre (TNFC) is an interdisciplinary research and service centre at the University of Toronto.

The centre offers global research leadership by providing access to state-of-the-art nanofabrication facilities, collaborative research networks, advanced educational opportunities and information exchange events for registered users and clients.



Key Research Areas

Blah .

Microelectronics	Microfluidics	MEMS/NEMS	Photonics	Plasmonics
CMOS Processing	Photovoltaics	Integrated Optics	Nanomaterial Processing	

Facilities



Pratt Microfabrication Facility

The Pratt Microfabrication Facility supports a wide range of silicon-based fabrication activities including photolithography, oxidation, thin film deposition, plasma etching, wet etching, device processing, packaging and analysis.

The 400m2 facility is located on the 4th floor of the Pratt engineering building at the University of Toronto, and includes both a class 10,000 (ISO Class 7) microfabrication processing area and a class 1,000 (ISO Class 6) photolithography processing area.



Wallberg Nanolithography Facility

The Wallberg Nanolithography Facility (an investment of over \$6 million) houses the state-of-the-art Raith EBP5000+ Electron Beam Lithography System. The 108 m2 facility is located in the basement of the Wallberg building at the University of Toronto in order to achieve very low mechanical vibration of <10 µm/s.

To ensure exceptional performance, the system is enclosed in an environmental chamber maintaining temperature stability of 21 ± 0.25 °C, 0.1 °C/hr. The 58 m2 Class 100 cleanroom lab space consists of the environmental chamber, control room, processing lab and gowning room, and contains a complete suite of equipment to perform nanolithography from CAD design to exposed resist development and analysis.

Capabilities

Electron Beam Lithography	+	Etching	+
Photolithography	+	Dicing & Packaging	+
Deposition	+	Imaging & Characterization	+

Services

User Access



All users must complete an [ECE User Access Form](#) and attend mandatory training sessions before using TNFC facilities.

All new and existing non-ECE staff members and students will be required to additionally fill out an [Attestation Form](#). Both forms are to be completed and sent to tnfc@utoronto.ca. Researchers will be charged for the use of the facilities on a per-minute basis as logged by our LMACS Access System, and invoices will be distributed on a quarterly basis.

For non-UofT academic and industrial researchers, please send an email to tnfc@utoronto.ca to work out an arrangement to come onsite for a short-term visit.

Contact TNFC

Learn More

Client Services



TNFC offers a wide range of fabrication and characterization services to academic and industrial clients. Many researchers require infrequent access to TNFC capabilities in order to prototype or analyze a small number of devices. For these cases, user access and training may not be practical. TNFC technical staff offer experienced service to perform fabrication and characterization processes on behalf of the user for a predetermined fee.

TNFC is also a proud member of the CMC FACT Network, which leverages experts and state-of-the-art toolsets from micro/nano fabrication laboratories at Canadian universities for custom fabrication and characterization services.

To request a service or for more information:

Contact TNFC

History

The Nortel Institute for Telecommunications (NIT) was originally established in 1997 with funding from the Nortel Networks Corporation. The institute included 10 open-research labs across the UofT campus.

In 1999, the institute received funding from Canada Foundation for Innovation (CFI), Ontario Research and Development Challenge Fund (ORDCF) and the Ontario Innovation Trust (OIT), which collectively funded the design and construction of the Prototyping Cleanroom in the Bahen Building. The cleanroom officially opened in May 2003. The institute was re-named the Emerging Communications Technology Institute (ECTI) in 2005, after the amalgamation of the Nortel Institute and the Micro Fabrication Research Laboratory in the Pratt Building.

In 2005, CFI funding obtained to construct an Electron Beam Nanolithography Facility in the basement of the Wallberg Building, as well as purchase a state-of-the-art Electron Beam Lithography tool. The Wallberg Nanolithography Facility was officially available to academic and industrial users in September 2009.

The institute was re-branded as the Toronto Nanofabrication Centre (TNFC) in April 2013.

Our Partners

Granting Agencies	Partner Organizations	Canadian Nanofabrication Facilities
   	  	   

Contact Us

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